

ICU Patient Phenotypes from MIMIC-III

Grouping ICU patients two different ways - by how sick they look right now, and by what chronic conditions they came in with - then checking whether the two groupings agree.

Why Look at This?

Doctors already do this

Clinicians informally sort patients into "the sick ones" and "the stable ones," and separately track who has a long medical history. We rarely check whether those two mental categories actually line up.

No labels required

Clustering doesn't need a known outcome to work from - it just groups patients who look similar to each other. That's different from a model trained to predict a specific label.

Something we can test

MIMIC-III lets us build both views of the same patients and directly check whether acute severity and chronic burden move together, or apart.

Setting Up the Problem

The Cohort

One row = one ICU stay. We use each patient's first ICU stay only, so a patient who came back several times doesn't skew the clustering toward their pattern alone.

Other cohort rules: adults only (18+), stay of at least 24 hours (we need a full first-day window), and newborn admissions excluded, since they behave very differently from adult stays.

Data: MIMIC-III, pulled through Google BigQuery.

Two Feature Sets, One Cohort

We build two separate descriptions of the same patients:

1. Vitals & labs (first 24h) - how sick the patient looks right now.
2. Comorbidities (diagnosis codes) - what chronic conditions the patient came in with.

Each gets its own clustering. Then we check whether the two clusterings agree.

What We Feed Each Clustering

Vitals & Labs

Heart rate, blood pressure, respiratory rate, temperature, SpO2, and GCS (consciousness score), plus creatinine, WBC, lactate, glucose, bicarbonate, potassium, and hemoglobin.

Each one is measured many times in the first 24 hours, so we summarize it as mean, min, max, and std (how much it fluctuated) - 49 columns in total before reduction.

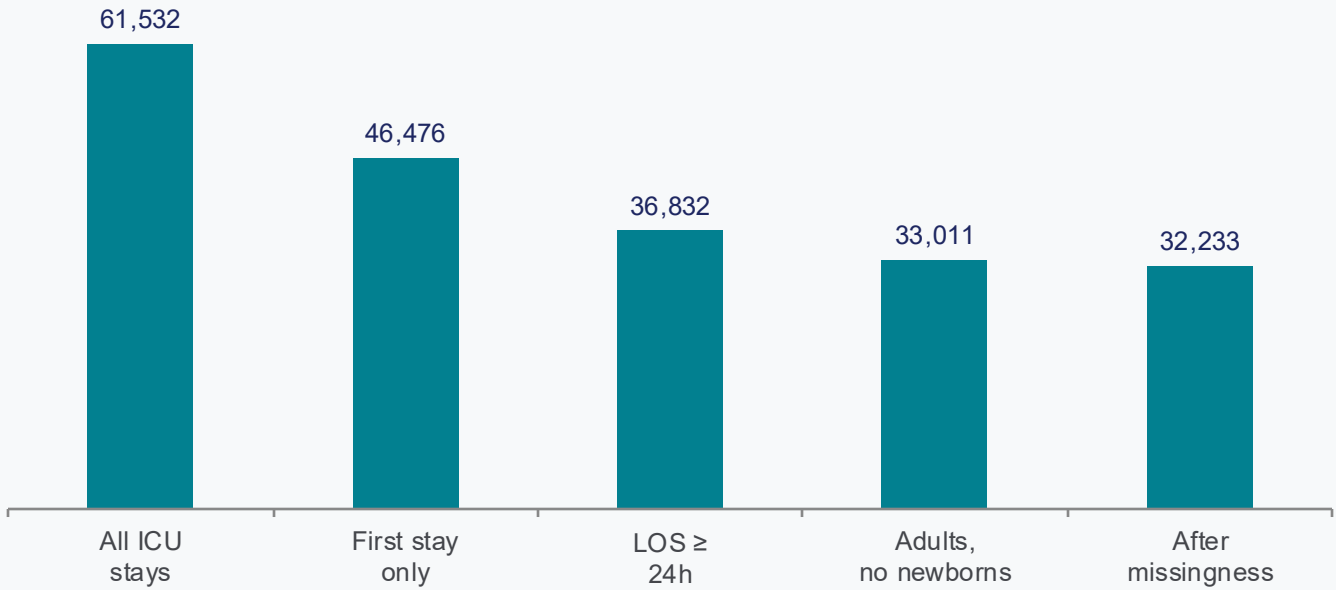
FINDING

MIMIC-III occasionally contains charting errors - a misplaced decimal can turn a respiratory rate of 23.5 into 2,355,555. Before any averaging happens, every reading is checked against a plausible physiological range (e.g. 0-80 breaths/min) and dropped if it falls outside it, so one bad entry can't distort a patient's whole feature row.

Comorbidities come from diagnosis codes (ICD-9), grouped into 15 chronic-condition categories - each one a simple yes/no flag.

Where the Cohort Came From

Cohort Funnel



FINDING

10.8%

of ICU stays in the final cohort ended in in-hospital death.

Getting the Data Model-Ready

1. Filter implausible readings

Every vital/lab reading is checked against a plausible clinical range before it's ever averaged, so one bad entry can't distort a patient's whole row.

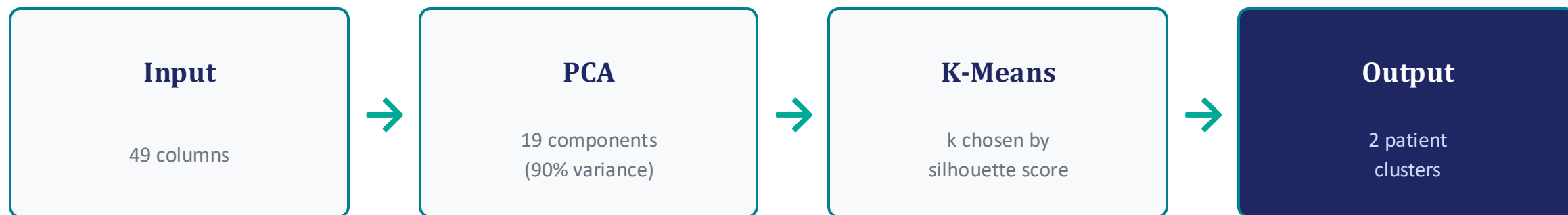
2. Impute + scale

Missing vitals/labs are filled with the median, then every column is standardized (mean 0, std 1) so no single measurement dominates just because of its natural scale.

3. Reduce with PCA

Mean/min/max/std of the same vital are highly correlated. PCA collapses 49 columns to 19 components that still capture 90% of the variance, without the redundancy.

From Raw Vitals to Clusters



Why PCA before clustering: heart_rate_mean, _min, _max, and _std are four separate columns describing one signal. Without reducing them first, vitals with more derived columns would quietly get more influence over which patients end up in the same cluster.

RESULT 1

Cluster on Vitals & Labs

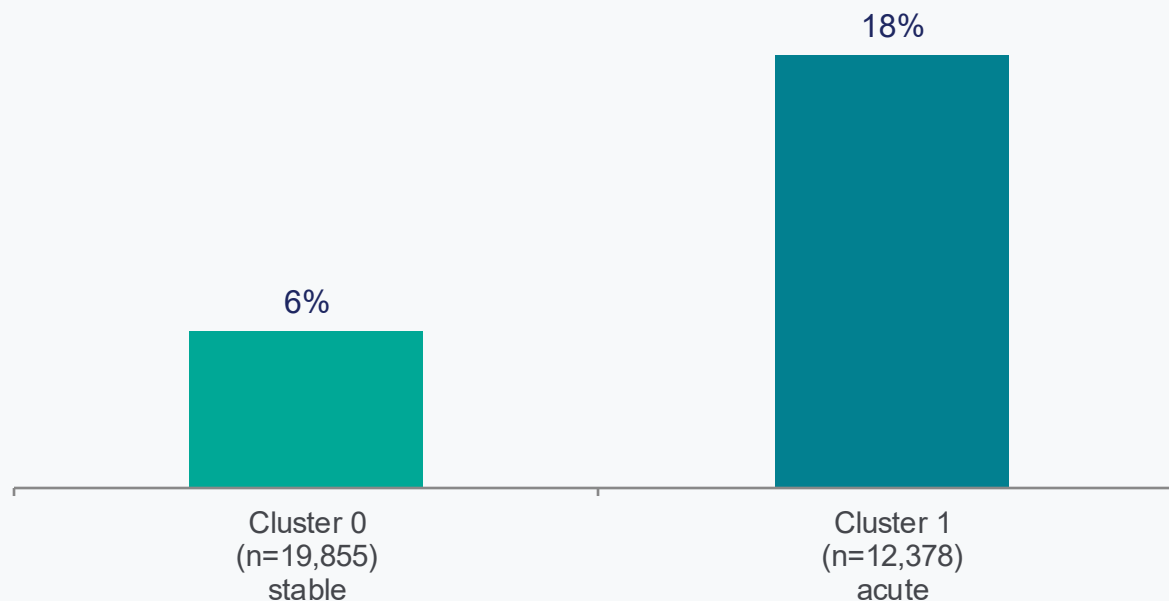
FINDING

k = 2

silhouette score = 0.121

Chosen by silhouette score - the highest of k=2 through k=7 tried. A modest score, meaning real but overlapping structure, not a dramatic split.

Mortality Rate by Cluster



RESULT 2

Cluster on Comorbidities

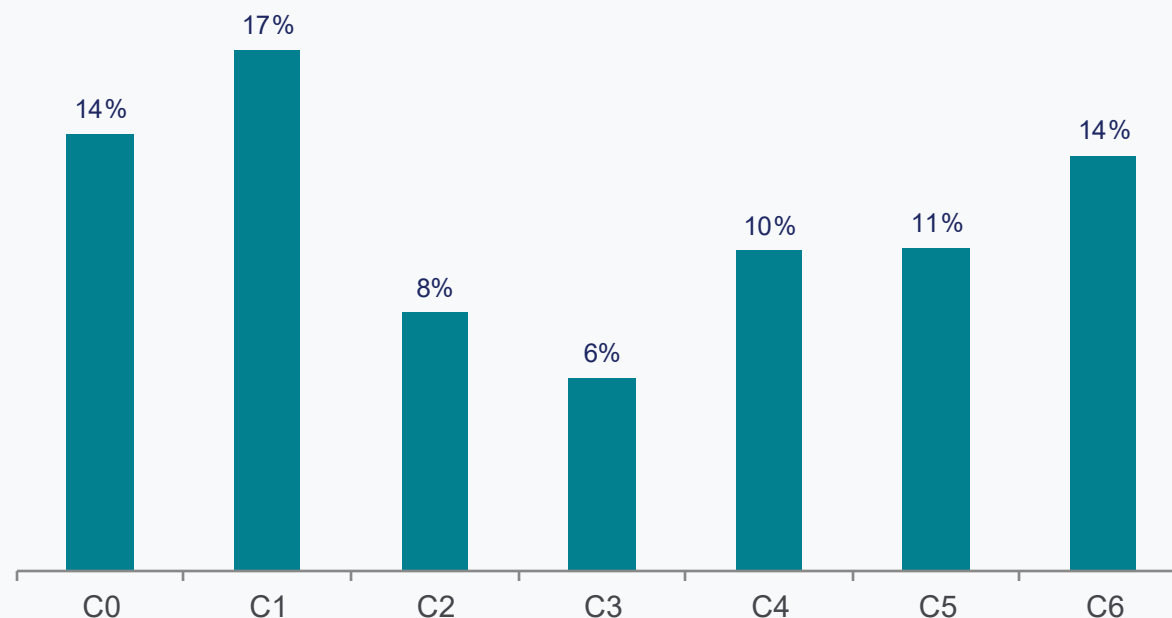
FINDING

k = 7

silhouette score = 0.196

Chosen by silhouette score - slightly higher than the vitals clustering, since binary yes/no combinations tend to separate a bit more crisply than overlapping continuous vitals.

Mortality Rate by Cluster



C6 = high CHF, diabetes, renal failure | C4 = cardiac arrhythmia + hypertension | C3 = lowest mortality, mostly anemia + hypertension

Why We Don't Trust P-Values Alone

P-value

Tells you whether a difference is likely real. With 32,233 patients, almost any real difference becomes "significant," so a tiny p-value alone doesn't tell you how much the clusters actually differ.

Cramer's V

Measures how strongly cluster membership is associated with mortality, on a 0-1 scale. Roughly: 0.1 = small, 0.3 = medium, 0.5+ = large effect.

Eta-squared

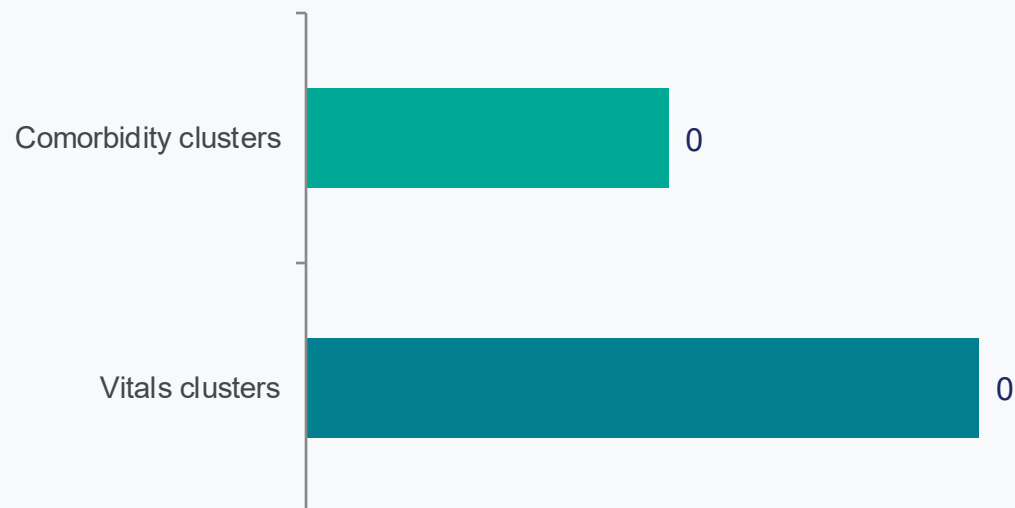
The share of variance in length of stay explained by which cluster a patient falls into, also 0-1. Same rough scale as Cramer's V.

RESULTS

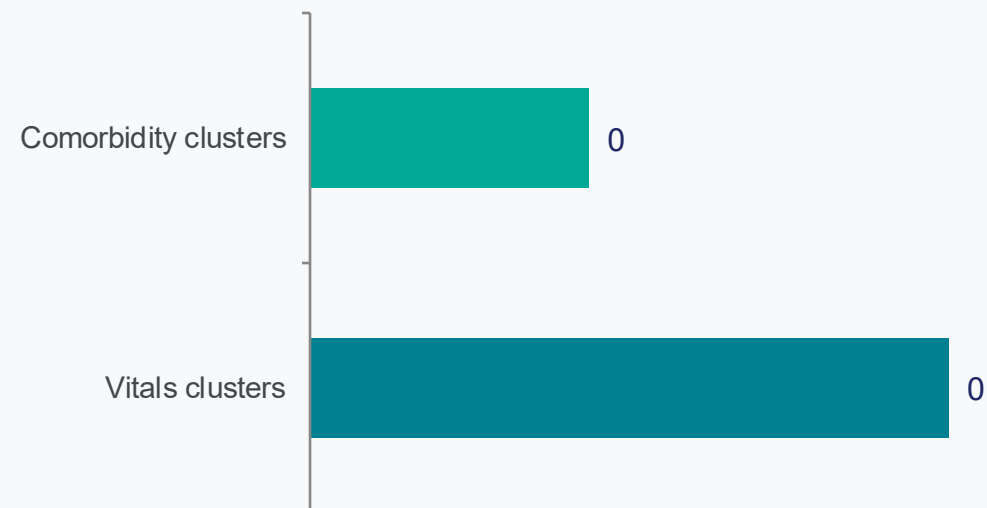
Effect Sizes: Vitals vs. Comorbidity Clusters

Both clusterings are statistically associated with outcomes ($p < 0.0001$ in every test) - but the effect sizes tell a more honest, more modest story.

Mortality (Cramer's V)



Length of Stay (eta-squared)



THE KEY QUESTION

Do the Two Views Agree?

FINDING

ARI 0.012
NMI 0.008

Both scores run 0 to 1, where 1 means the two clusterings are identical and 0 means no better than random agreement. Close to zero means a patient's acute-vitals cluster tells you almost nothing about which comorbidity cluster they're in.

What This Means

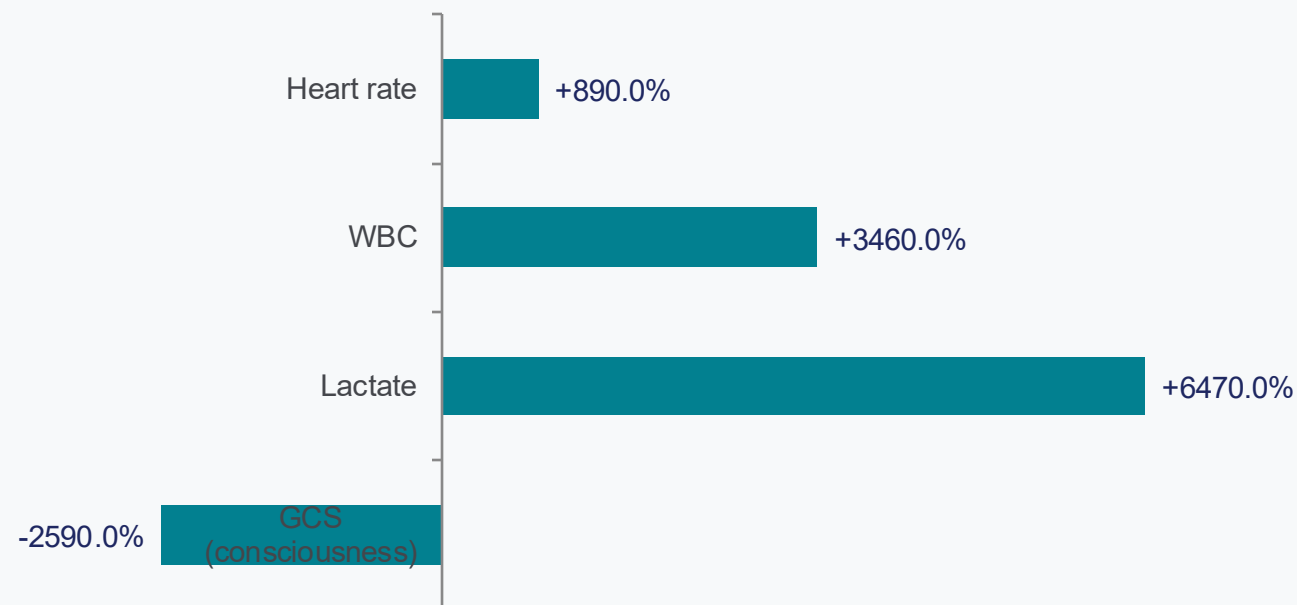
Acute severity and chronic burden behave like two largely independent axes in this cohort.

A patient can be chronically ill with few visible signs of acute distress right now - or acutely crashing with a light comorbidity history.

Neither view can substitute for the other.

What Actually Distinguishes the Clusters?

Cluster 1 ("acute") vs. Cluster 0 ("stable") - % difference



Averages: Cluster 0 - GCS 13.5, lactate 1.7, WBC 10.7, HR 82.8 | Cluster 1 - GCS 10.0, lactate 2.8, WBC 14.4, HR 90.2

FINDING

Cluster 1 has lower consciousness (GCS), higher lactate (poor tissue oxygenation), and higher white cell count - a coherent "sicker right now" signature the clustering found without ever being told the outcome.

What We Learned

Takeaways

- Acute vitals cluster into a real, if modest, “sicker vs. stable” split (silhouette 0.121).
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- Comorbidities cluster into 7 distinct chronic-disease profiles (silhouette 0.196).
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- The two clusterings barely agree with each other (ARI 0.012) - the main finding of this project.
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- Both associate with mortality/LOS, but with small effect sizes - real, not dominant.

Limitations

- Comorbidity mapping is a simplified 15-category subset of Elixhauser, not the full validated index.
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- Vitals only cover the first 24 hours - a patient who crashes on day 3 looks the same as one who stayed stable.
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- K-Means isn't the ideal method for binary comorbidity data - used for both sets for direct comparability.
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- No comparison against other clustering algorithms (GMM, hierarchical) or hyperparameter search.

Try It Yourself

- 1 Get PhysioNet-credentialed access to MIMIC-III, and load it into Google BigQuery.
- 2 Run the SQL queries in the notebook to build the cohort, vitals/labs, and comorbidity tables.
- 3 Install scikit-learn, scipy, and umap-learn.
- 4 Run the notebook top to bottom - no manual steps needed in between.